SDITO-LAB

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Assignment NO : 10

Problem Statement : Deploy a project from GitHub to EC2 by creating a new Security Group and configuring User Data.

AIM

To launch an EC2 instance, create and configure a new Security Group, deploy a project from GitHub using User Data script, and access the deployed application via public IP.

Prerequisites

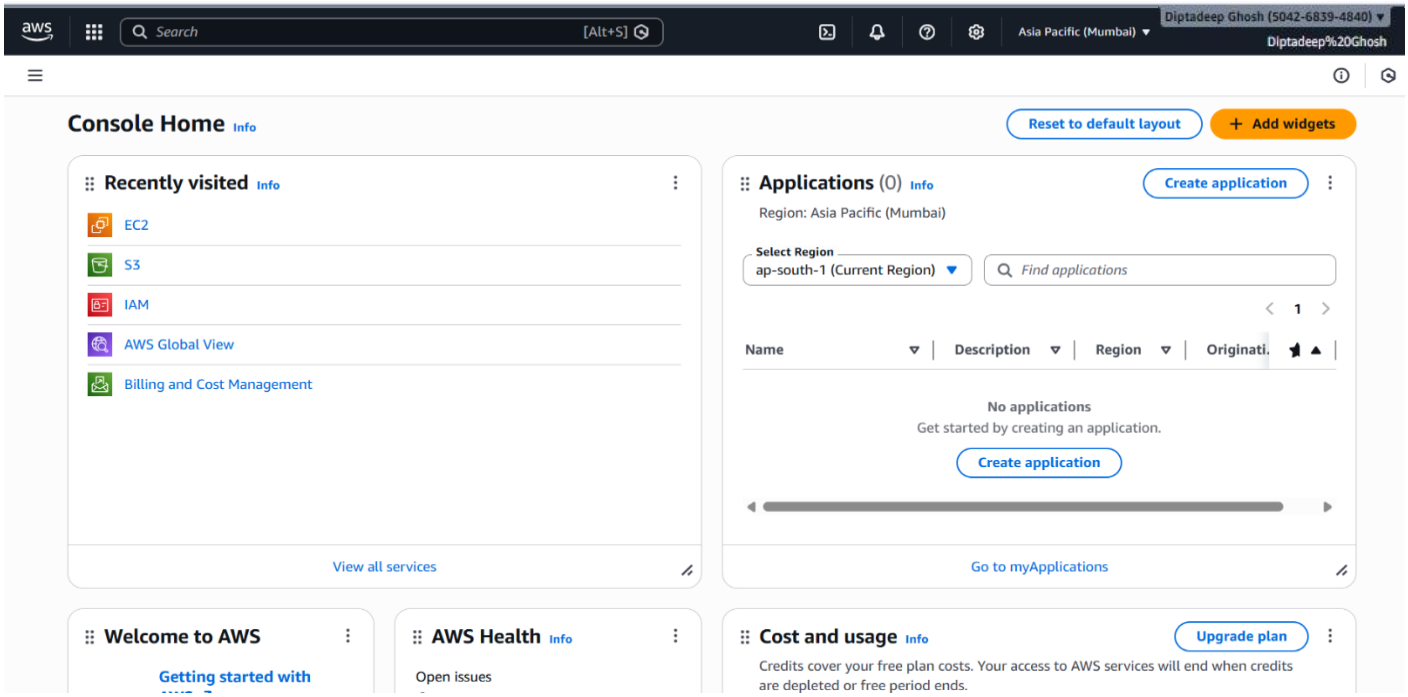
- AWS Account
- GitHub repository (public)
- Basic familiarity with Linux shell command, Git , and Node.js.
- Key Pair for EC2 login
- An AWS account with permission to create EC2 instances, security groups, and use EC2 Instance Connect.

Solution:

PHASE 1 – Create a Security Group

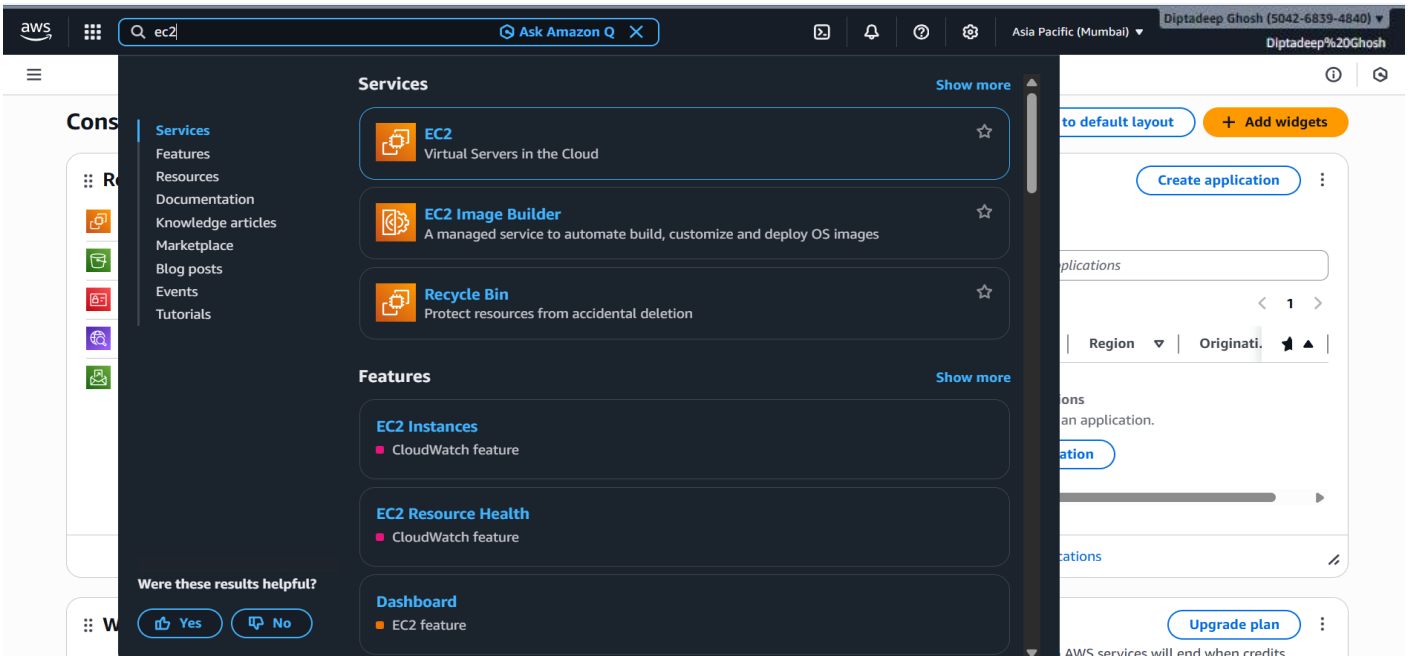
Step 1: Login to AWS Console

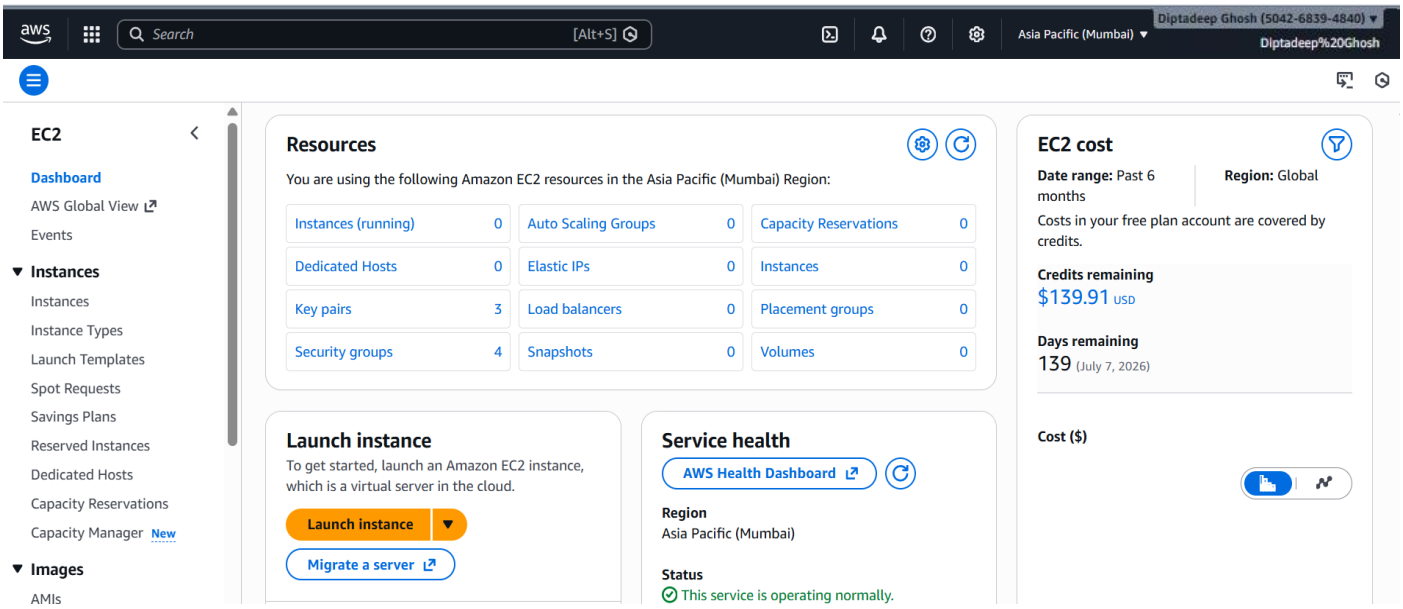
1. Login to AWS Management Console.

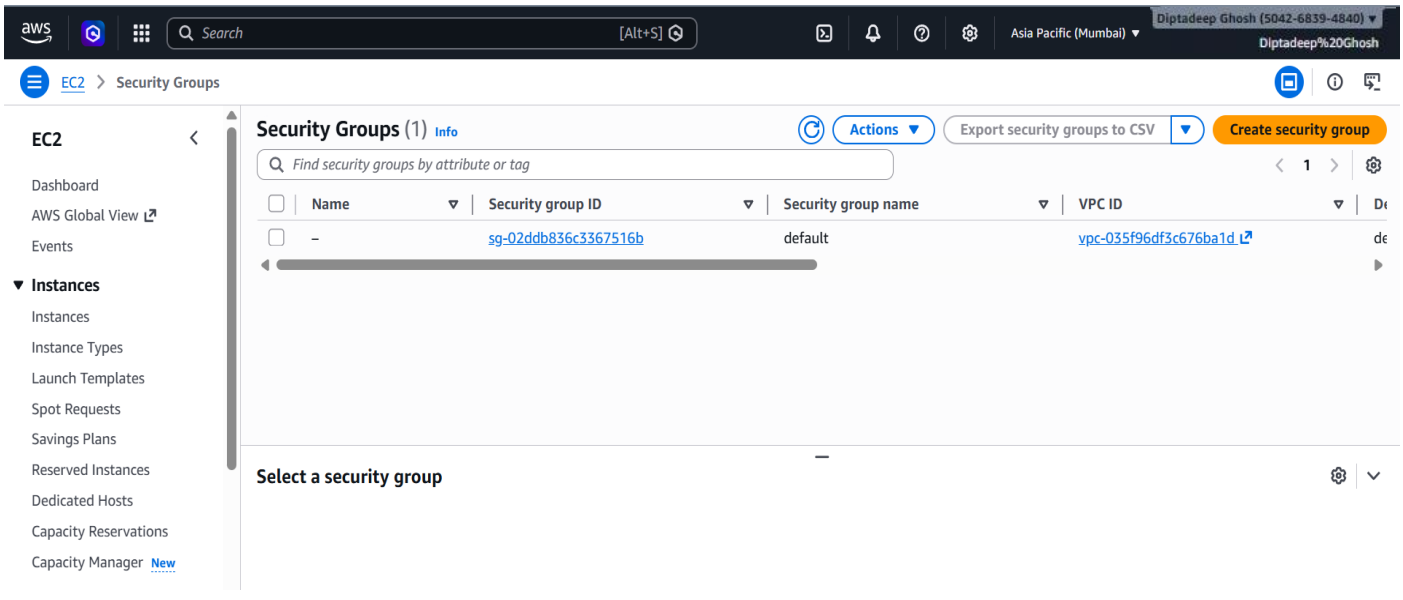


Step 2: Create a New Security Group

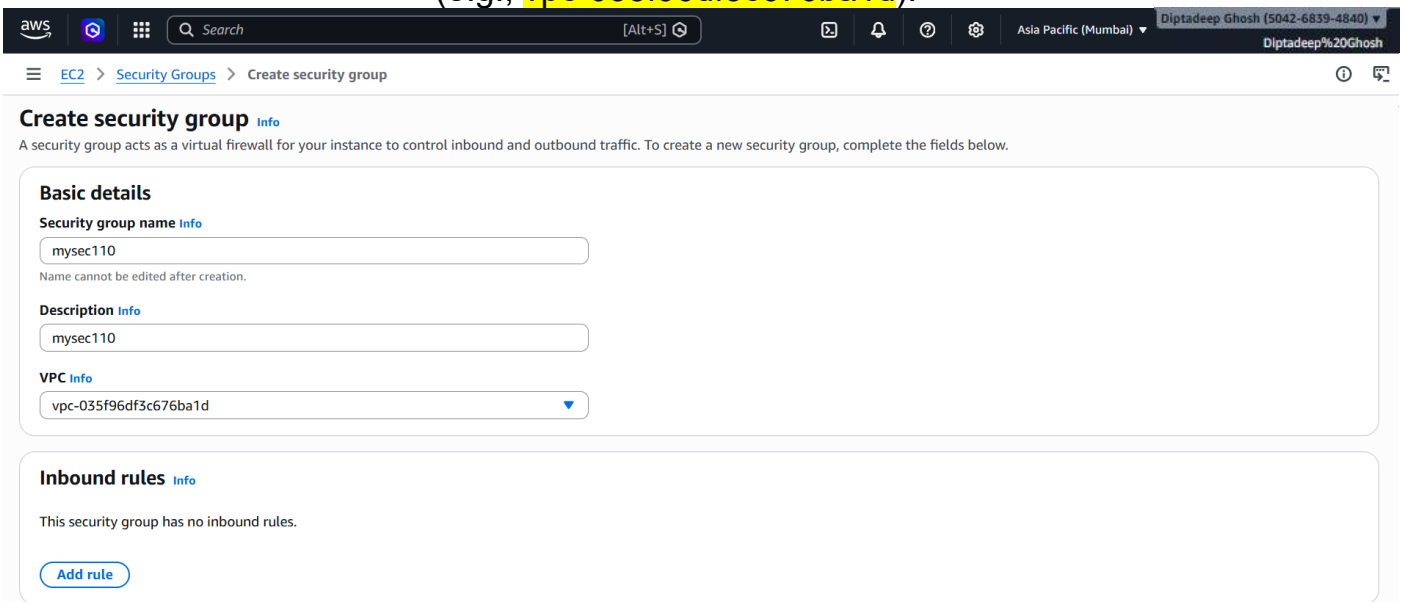
1. In the AWS Console, go to **EC2** > **Network & Security** > **Security Groups**.





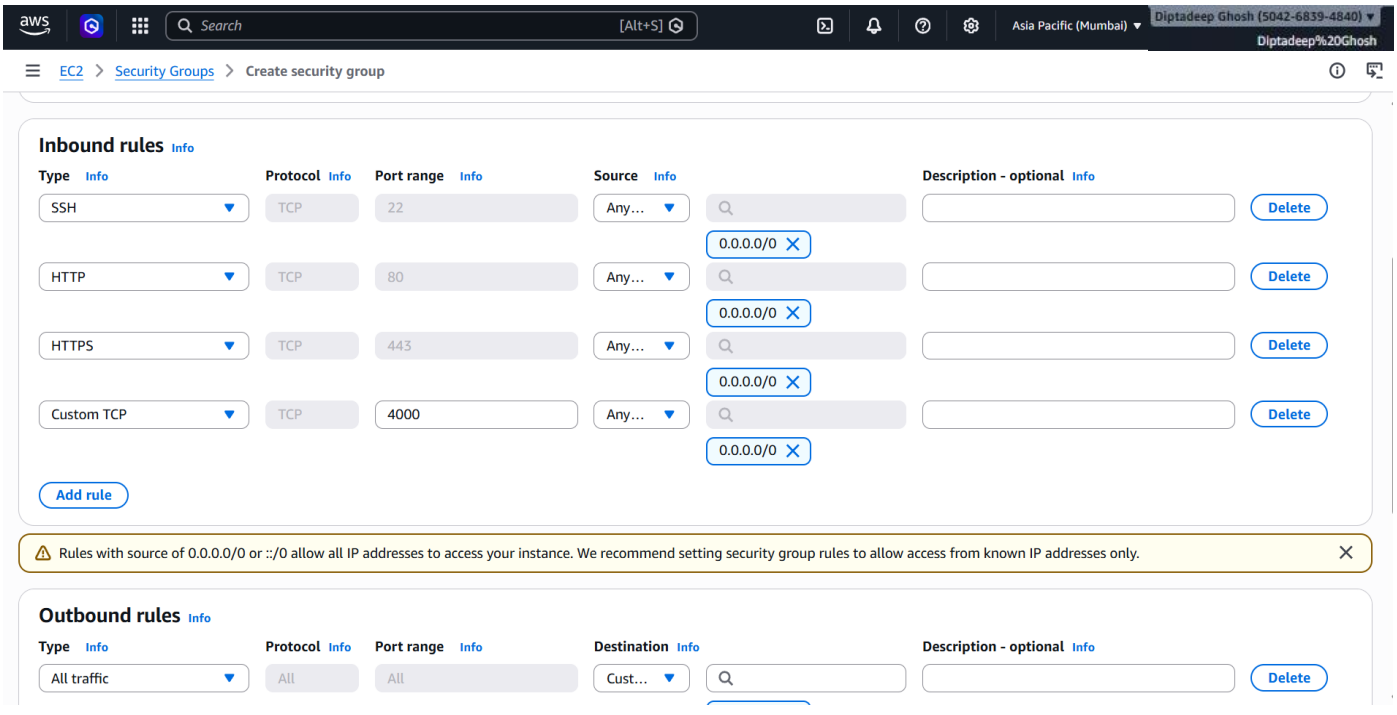


2. Click **Create security group**.
3. Fill basic details:
    - **Security group name:** mysec110
    - **Description:** mysec110
    - **VPC:** select the VPC where you will launch the instance (e.g., vpc-035f96df3c676ba1d).

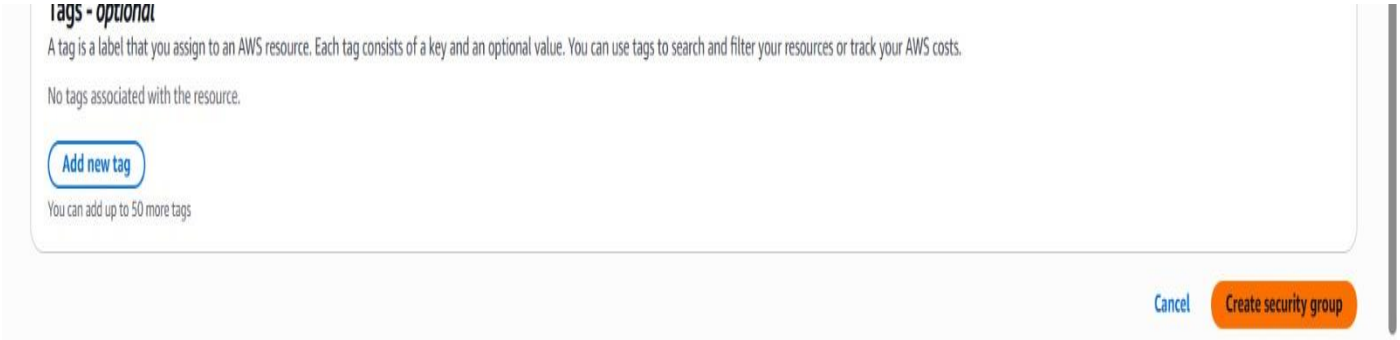


4. Click on **Add Inbound Rule** and **configure**.

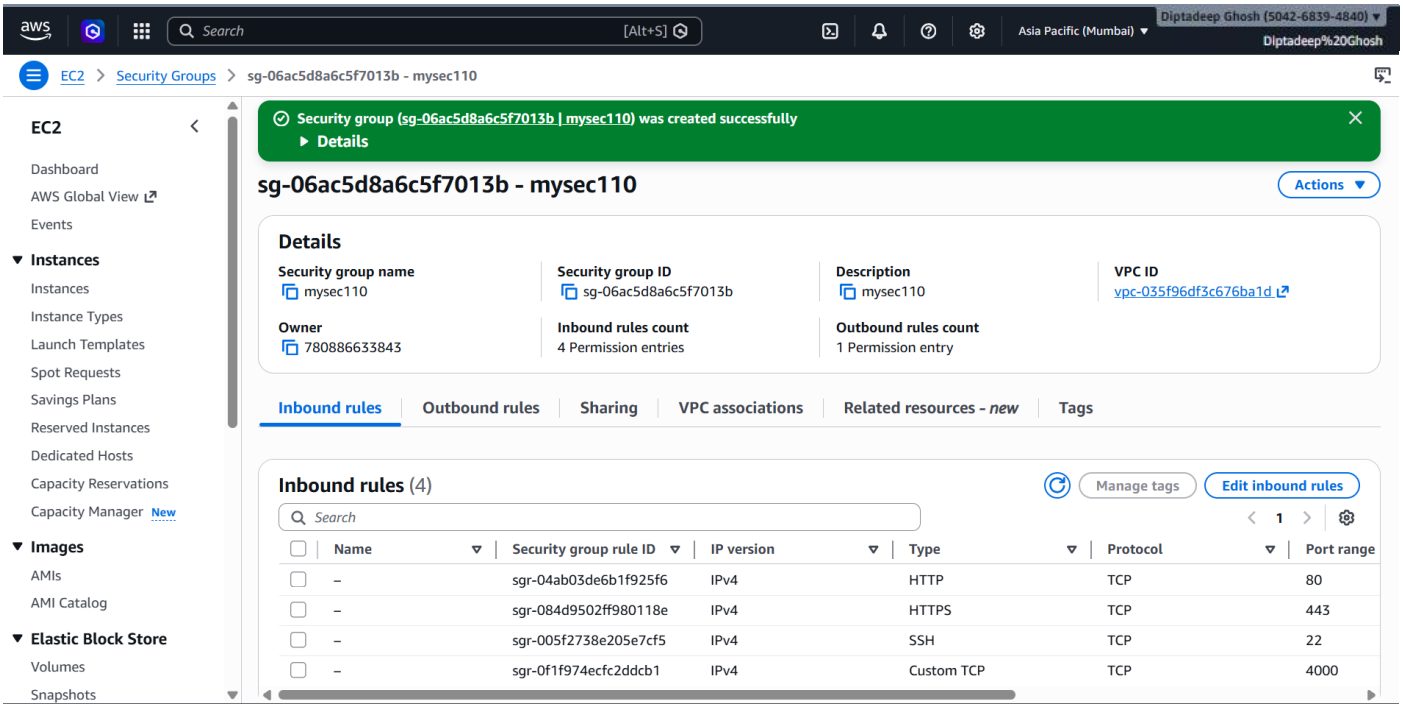
- Add **SSH** (Type: **SSH**, Protocol: **TCP**, Port: **22**) — Source: **anywhere-ipv4** (0.0.0.0/0)
- Add **HTTP** (Type: **HTTP**, **TCP**, Port: **80**) — Source: **anywhere-ipv4** (0.0.0.0/0).
- Add **HTTPS** (Type: **HTTPS**, **TCP**, Port: **443**) — Source: **anywhere-ipv4** (0.0.0.0/0).
- Add **Custom TCP** (Type: **Custom TCP**, Protocol: **TCP**, Port range: **4000**) — Source: **anywhere-ipv4** (0.0.0.0/0).

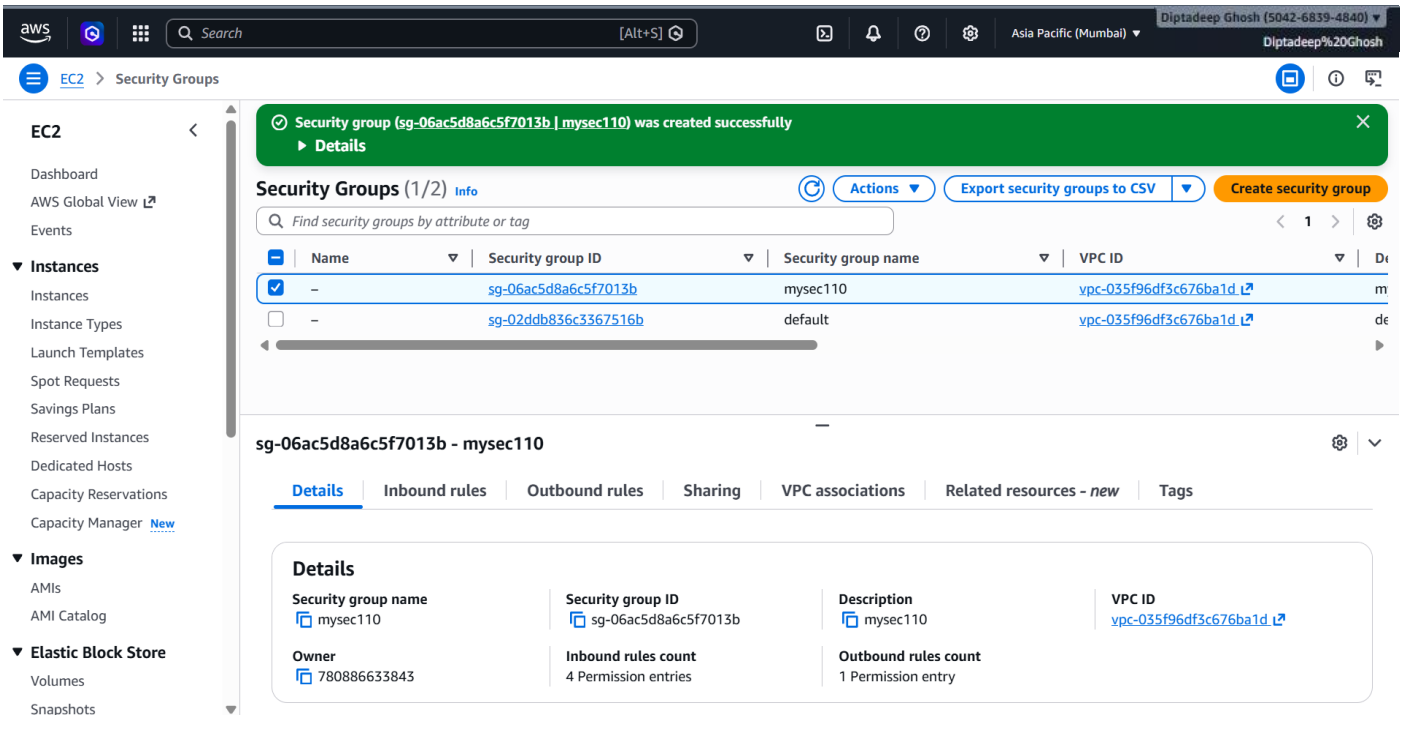


5. Configure **Outbound rules**: default is *All traffic* to 0.0.0.0/0. Leave as it is.
6. Now , Click on **Create security group**.



- Now we can see a success message and the new **mysec110** security group is listed.

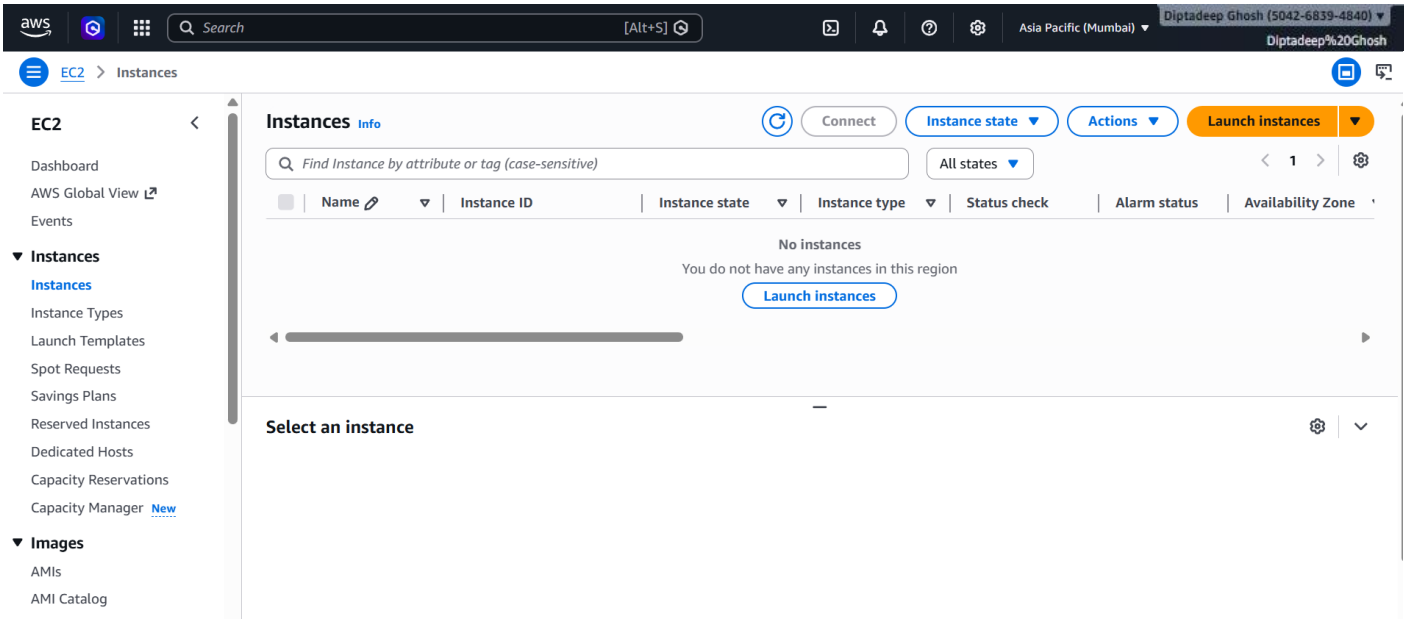




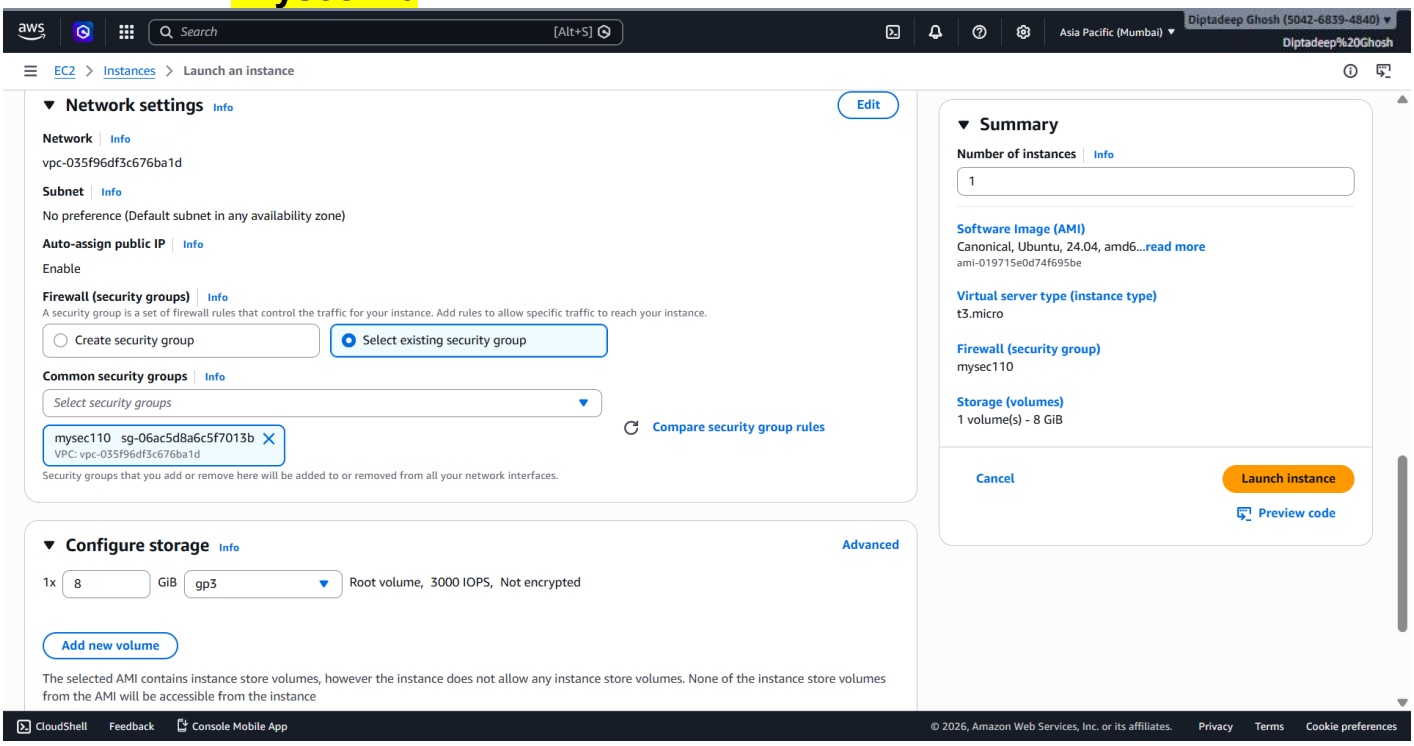
PHASE 2 – Launch an EC2 Instance and Add User Data Script

Step 1: Launch an EC2 Instance and Attach the Security Group

1. Go to **EC2** > **Instances** > **Launch instances**.



2. Choose an AMI: ***Canonical, Ubuntu 24.04*** .
3. Choose Instance Type: **t3.micro** (free-tier account) or another small type.
4. Create a key Pair in my case : **mysec110.**
5. Configure instance details:
    - **Network (VPC):** choose the **same VPC** used for the security group.
    - **Subnet:** choice **default**.
    - **Auto-assign public IP: Enable**
6. **Firewall (security groups):** choose **Select existing security group**
7. **Common Security Group :** Select the newly created Security Group. In my case its - **mysec110**.



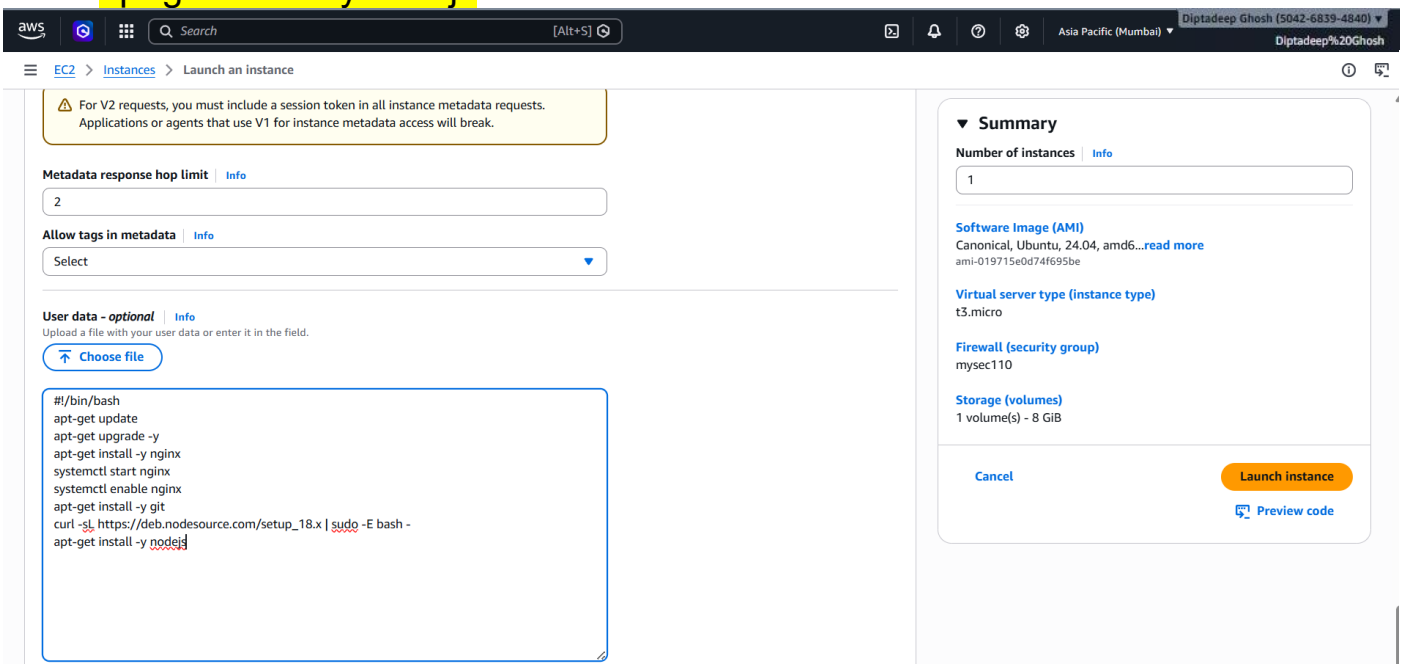
8. Scroll to **'Advanced Details'** section.

Step 2: Add User Data Script

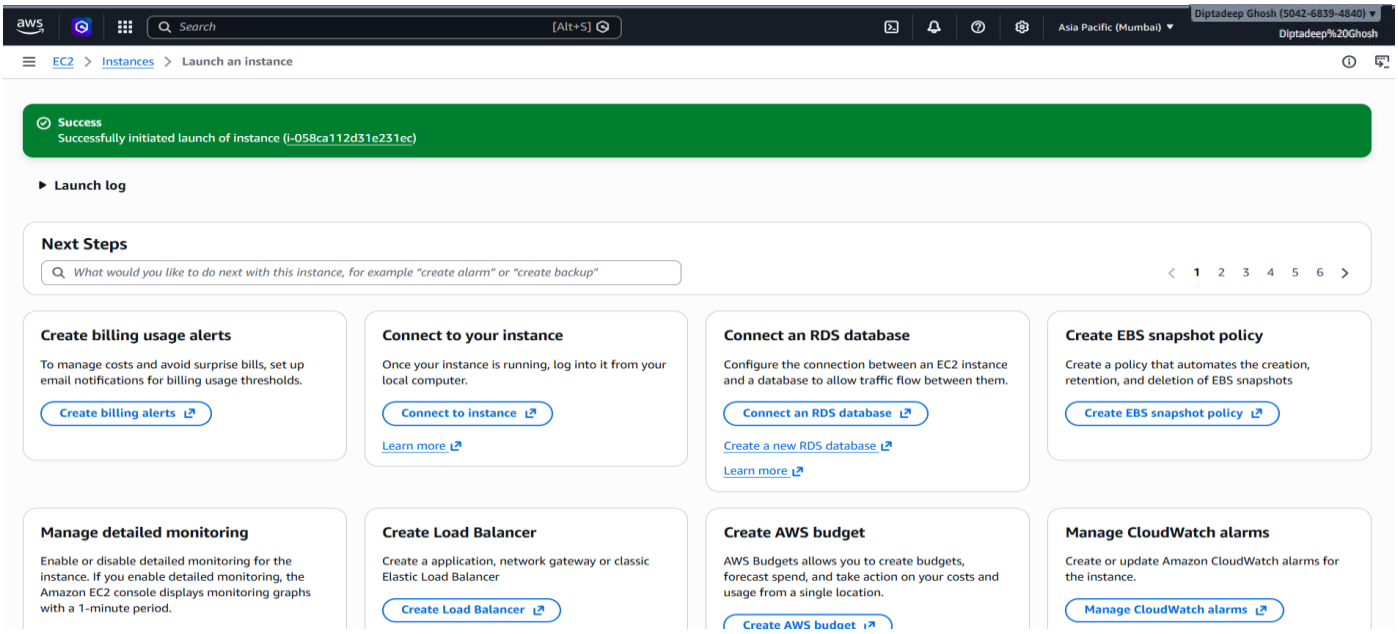
1. Click on **Advance Details** .
2. **User data:** In that, paste the following script: (below). This runs on first boot and prepares the instance.

paste in the *User data* box:

```
#!/bin/bash
apt-get update
apt-get upgrade -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
```



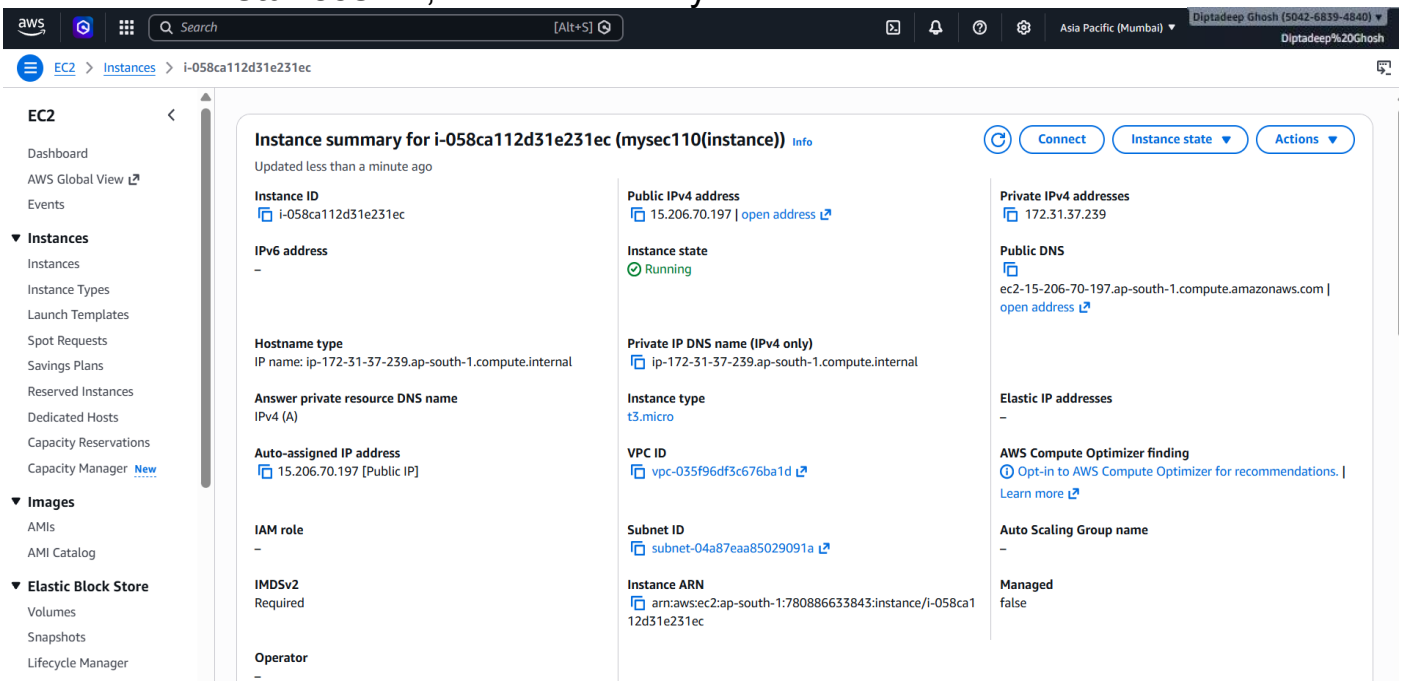
3. Click '**Launch instance**'.



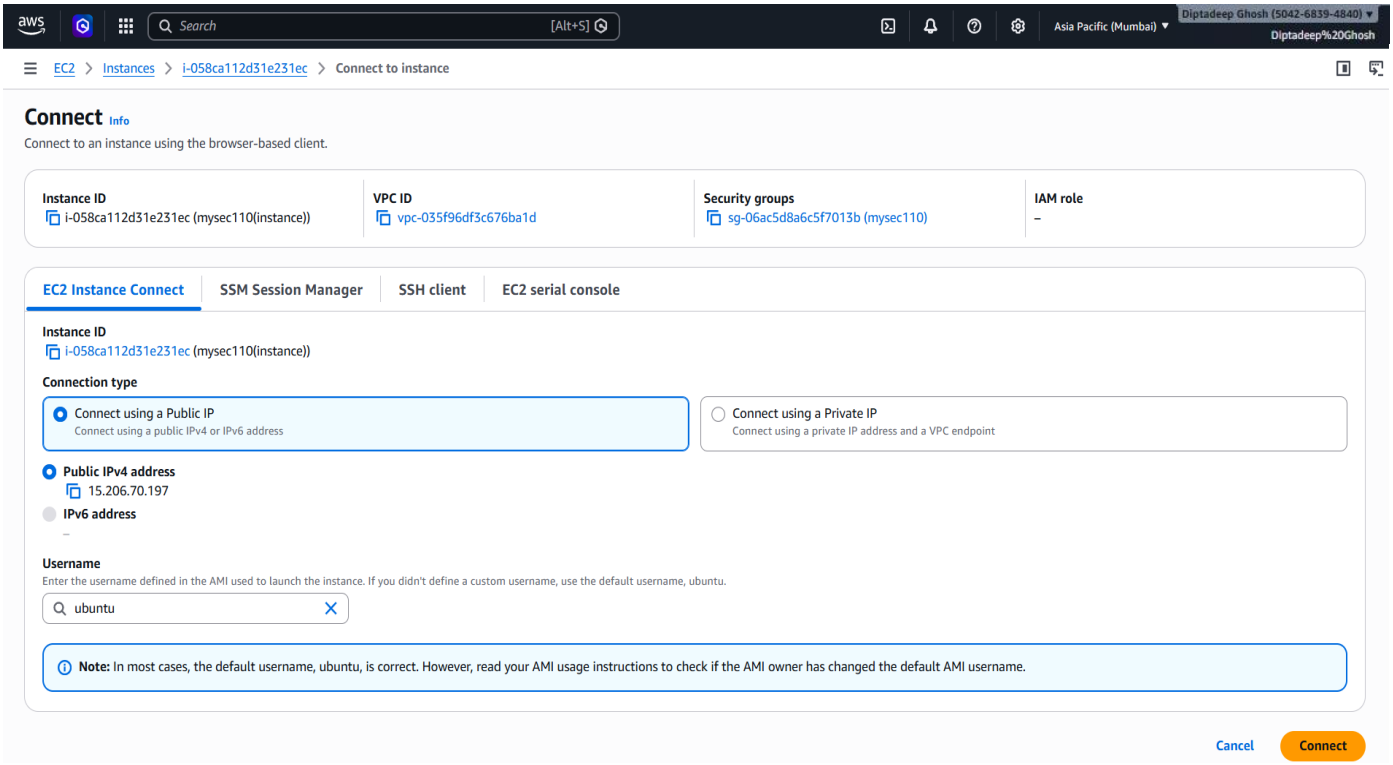
PHASE 3 – Connect to the Instance (EC2 Instance Connect)

Step 1: Connect to EC2 Instance

1. In the **Instances** list, select the newly launched instance.



2. Click **Connect** > **EC2 Instance Connect**.
3. Choose **Connect using a Public IP** (default). The **Username** for the Ubuntu AMI is **ubuntu**.
4. Click **Connect** — It will open a browser-based terminal session.

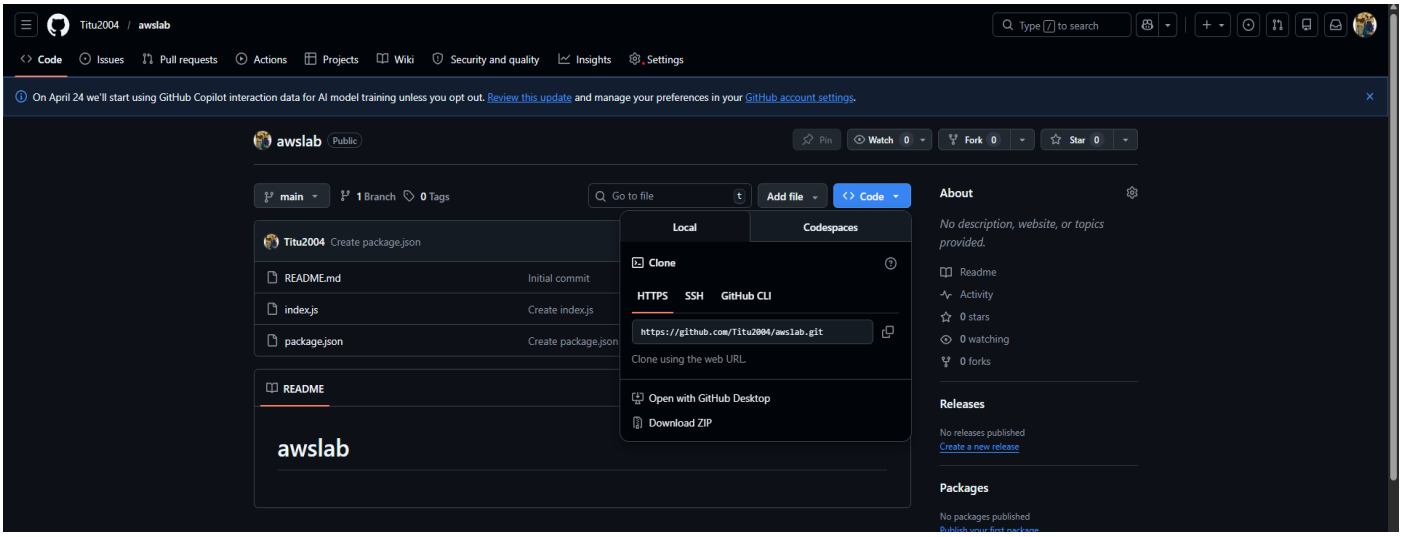


*(Then Its show successful connection and the **public IP: 15.206.70.197** and username **ubuntu**.)*

PHASE 4 – Pull the Project from GitHub and Start the Website

Step 1: Clone the REPO From GITHUB

1. Go to **GITHUB Account** .
2. Open the which have node files , in my case 'awslab' , and **copy** the REPO **URL Link**.



3. Then, Once connected via Instance Connect, **perform these commands** in the **terminal:**

For clone the GitHub repo used
git clone https://github.com/Titu2004/awslab.git

To change the directory & for checking the files that clone actually done
cd aws
ls

It will show
index.js package.json

To Install Node dependencies
npm install

To Start the Node app
node index.js

It will show Result as: **"Started server"**

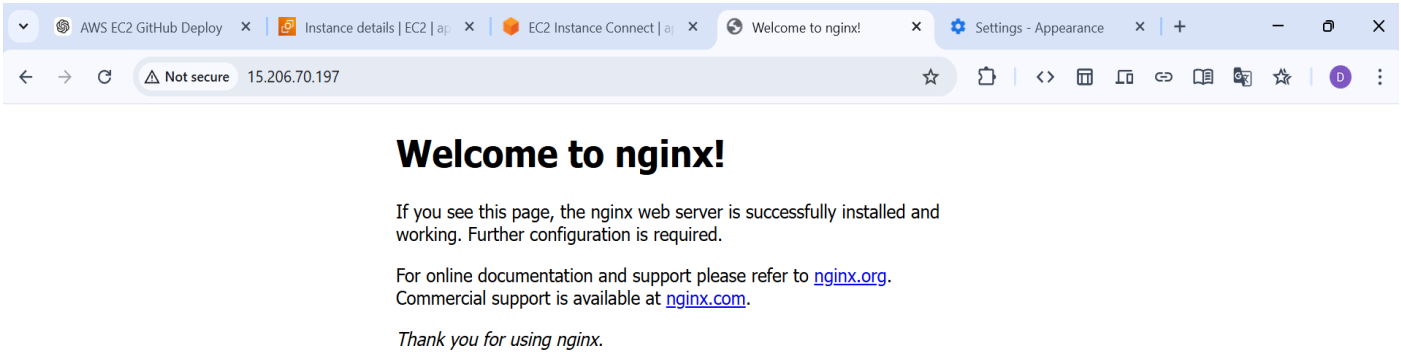
```
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 15 (delta 5), reused 4 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), 4.15 KiB | 1.38 MiB/s, done.
Resolving deltas: 100% (5/5), done.
ubuntu@ip-172-31-37-239:~$ cd awslab
ubuntu@ip-172-31-37-239:~/awslab$ ls
index.js  package.json
ubuntu@ip-172-31-37-239:~/awslab$ npm install

added 224 packages, and audited 225 packages in 23s

26 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.10.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.10.1
npm notice To update run: npm install -g npm@11.10.1
npm notice
ubuntu@ip-172-31-37-239:~/awslab$ node index.js
Started server
```

i-058ca112d31e231ec (mysec110(instance))

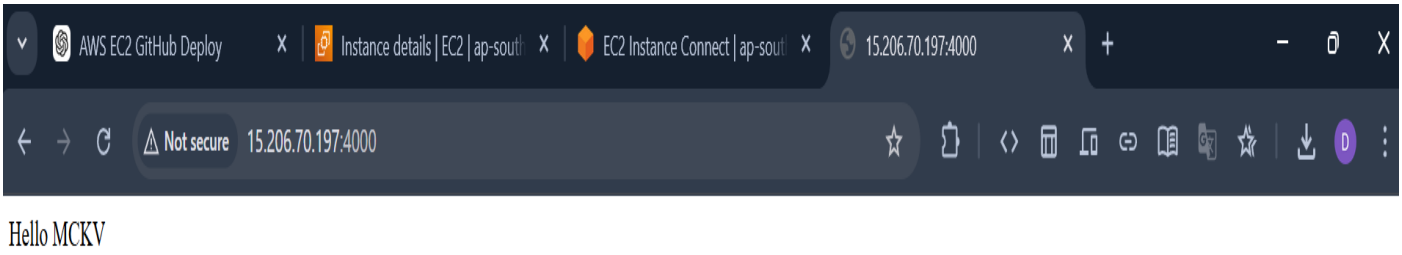
PublicIPs: 15.206.70.197 PrivateIPs: 172.31.37.239

Step 2: Verify This From MY Browser

1. Copy the **Public IPv4 address** from **EC2 Instance dashboard**.
2. Check the default nginx page (HTTP port 80): open **http://<PUBLIC_IP>** (e.g., http://15.206.70.197) — we can see **Welcome to nginx!**



3. Check the Node.js app on port 4000: open **http://<PUBLIC_IP>:4000** (e.g., http://15.206.70.197:4000)— we can see the web output -→ **Hello MCKV**



Output :

The project hosted on GitHub is successfully deployed on AWS

EC2 instance and accessible via public IP address.

Conclusion :

This lab demonstrates how to deploy a GitHub project on AWS EC2 by configuring a Security Group and using User Data automation script.